

Continuum Mechanics

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Chapter 06

Isothermal Fluid Mechanics Part 2



Irrotational Motion and Potential Flow in Ideal Fluids



Result 6.4 (Criterion for Irrotational Motion in Ideal Fluids)

Let $\boldsymbol{\phi} : B \times [0, \infty) \to \mathbb{E}^3$ be a motion of an ideal fluid with spatial velocity $\mathbf{v}$ and constant mass density ρ_0. Suppose the body force is conservative, $\mathbf{b} = -\nabla^{\mathrm{x}}\beta$. If $\mathbf{v}$ is irrotational at $t = 0$, then it is irrotational for every $t \geq 0$.



Proof.

Let $\mathbf{w} = \nabla^{\mathrm{x}} \times \mathbf{v}$. Recall the identity from Result 6.3:

$$\dot{\mathbf{v}} = \frac{\partial \mathbf{v}}{\partial t} + \frac{1}{2}\nabla^{\mathrm{x}}|\mathbf{v}|^2 + (\nabla^{\mathrm{x}} \times \mathbf{v}) \times \mathbf{v}. \tag{6.5_2}$$

The momentum equation is

$$\rho_0 \dot{\mathbf{v}} = -\nabla^{\mathrm{x}} p - \rho_0 \nabla^{\mathrm{x}} \beta.$$

Taking its curl, using (6.5_2) and the fact that the curl of a gradient is zero, gives

$$\rho_0 \left[\frac{\partial \mathbf{w}}{\partial t} + \nabla^{\mathrm{x}} \times (\mathbf{w} \times \mathbf{v}) \right] = \mathbf{0}. \tag{6.9}$$

By the assumption on $\mathbf{v}$, $\mathbf{w}(\mathbf{x}, 0) = \mathbf{0}$.

Proof (Cont.)

Expanding the curl in (6.9) and dividing by $\rho_0 > 0$ gives

$$\frac{\partial \mathbf{w}}{\partial t} + (\nabla^{\mathrm{x}}\mathbf{w})\mathbf{v} - (\nabla^{\mathrm{x}}\mathbf{v})\mathbf{w} + \mathbf{w}(\nabla^{\mathrm{x}} \cdot \mathbf{v}) - \mathbf{v}(\nabla^{\mathrm{x}} \cdot \mathbf{w}) = \mathbf{0}.$$

Now $\nabla^{\mathrm{x}} \cdot \mathbf{v} = 0$ by incompressibility, and $\nabla^{\mathrm{x}} \cdot \mathbf{w} = 0$ because the divergence of a curl is zero. Hence

$$\dot{\mathbf{w}} - (\nabla^{\mathrm{x}}\mathbf{v})\mathbf{w} = \mathbf{0}, \tag{6.10}$$

where $\dot{\mathbf{w}} = \partial \mathbf{w}/\partial t + (\nabla^{\mathrm{x}}\mathbf{w})\mathbf{v}$.

We next show directly that $\mathbf{w}(\mathbf{x}, t) = \mathbf{0}$ is the unique solution of (6.10) with $\mathbf{w}(\mathbf{x}, 0) = \mathbf{0}$, using a change of variables.

Proof (Cont.)

Let $\mathbf{w}_m$ be the material description of $\mathbf{w}$:

$$\mathbf{w}_m(\mathbf{X}, t) = \mathbf{w}(\boldsymbol{\phi}(\mathbf{X}, t), t).$$

Define the material field $\boldsymbol{\xi}$ by

$$\boldsymbol{\xi} = F^{-1}\mathbf{w}_m, \qquad \text{equivalently} \qquad \mathbf{w}_m = F\boldsymbol{\xi}. \tag{6.11}$$

Differentiating the second equality gives

$$\frac{\partial \mathbf{w}_m}{\partial t} = \left(\frac{\partial F}{\partial t}\right)\boldsymbol{\xi} + F\left(\frac{\partial \boldsymbol{\xi}}{\partial t}\right).$$

Solving for $\partial\boldsymbol{\xi}/\partial t$ and eliminating $\boldsymbol{\xi}$, we obtain

$$\frac{\partial \boldsymbol{\xi}}{\partial t} = F^{-1}\left[\frac{\partial \mathbf{w}_m}{\partial t} - \left(\frac{\partial F}{\partial t}\right)F^{-1}\mathbf{w}_m\right]. \tag{6.12}$$

Proof (Cont.)

Recall the material-description identities

$$\frac{\partial \mathbf{w}_m}{\partial t} = (\dot{\mathbf{w}})_m, \qquad \left(\frac{\partial F}{\partial t}\right) F^{-1} = (\nabla^{\mathrm{x}} \mathbf{v})_m.$$

Thus the right-hand side of (6.12) vanishes by (6.10):

$$\begin{aligned} \frac{\partial \boldsymbol{\xi}}{\partial t} &= F^{-1} \left[(\dot{\mathbf{w}})_m - (\nabla^{\mathrm{x}} \mathbf{v})_m \mathbf{w}_m \right] \\ &= F^{-1} \left[\dot{\mathbf{w}} - (\nabla^{\mathrm{x}} \mathbf{v}) \mathbf{w} \right]_m = \mathbf{0}. \end{aligned}$$

Therefore, for all $\mathbf{X} \in B$ and $t \geq 0$,

$$\boldsymbol{\xi}(\mathbf{X}, t) = \boldsymbol{\xi}(\mathbf{X}, 0).$$

Moreover, $F(\mathbf{X}, 0) = I$, so (6.11) gives

$$\boldsymbol{\xi}(\mathbf{X}, 0) = \mathbf{w}_m(\mathbf{X}, 0).$$

Proof (Cont.)

Combining these facts with $\mathbf{w}_m = F\boldsymbol{\xi}$ in (6.11), we find

$$\begin{aligned}\mathbf{w}_m(\mathbf{X}, t) &= F(\mathbf{X}, t)\boldsymbol{\xi}(\mathbf{X}, t) \\ &= F(\mathbf{X}, t)\boldsymbol{\xi}(\mathbf{X}, 0) \\ &= F(\mathbf{X}, t)\mathbf{w}_m(\mathbf{X}, 0).\end{aligned}$$

Thus, if the motion is initially irrotational, then

$$\mathbf{w}_m(\mathbf{X}, 0) = \mathbf{0} \qquad \text{for all } \mathbf{X} \in B.$$

It follows that

$$\mathbf{w}_m(\mathbf{X}, t) = \mathbf{0} \qquad \text{for all } \mathbf{X} \in B, \quad t \geq 0.$$

Hence the motion is irrotational for all time. □

Result 6.5 (Bernoulli's Theorem for Ideal Fluids)

Let an ideal-fluid motion be irrotational with $\mathbf{v} = \nabla^{\mathrm{x}}\phi$, and suppose $\mathbf{b} = -\nabla^{\mathrm{x}}\beta$. Then balance of linear momentum implies

$$\nabla^{\mathrm{x}}\left(\frac{\partial\phi}{\partial t} + \frac{1}{2}|\mathbf{v}|^2 + \rho_0^{-1}p + \beta\right) = \mathbf{0}, \tag{6.13}$$

or, equivalently,

$$\frac{\partial\phi}{\partial t} + \frac{1}{2}|\mathbf{v}|^2 + \rho_0^{-1}p + \beta = f(t). \tag{6.14}$$



Proof.

Using Result 6.3, $\mathbf{v} = \nabla^{\mathrm{x}}\phi$, and $\mathbf{b} = -\nabla^{\mathrm{x}}\beta$, the momentum equation becomes

$$\rho_0 \left[\frac{\partial}{\partial t}(\nabla^{\mathrm{x}}\phi) + \frac{1}{2}\nabla^{\mathrm{x}}|\mathbf{v}|^2 \right] = -\nabla^{\mathrm{x}}p - \rho_0 \nabla^{\mathrm{x}}\beta.$$

Since $\partial/\partial t$ and ∇^{x} commute,

$$\nabla^{\mathrm{x}} \left(\frac{\partial \phi}{\partial t} + \frac{1}{2}|\mathbf{v}|^2 + \rho_0^{-1} p + \beta \right) = \mathbf{0}.$$

The quantity in parentheses is spatially constant on each connected component, so it equals a function $f(t)$. □

Potential-Flow Boundary-Value Problem



For an irrotational ideal fluid in a fixed region D,

$$\mathbf{v} = \nabla^{\mathrm{x}}\phi.$$

Incompressibility gives Laplace's equation

$$\Delta^{\mathrm{x}}\phi = 0, \qquad \mathbf{x} \in D, \quad t \geq 0, \tag{6.15}$$

and impermeability gives the homogeneous Neumann condition

$$\nabla^{\mathrm{x}}\phi \cdot \mathbf{n} = 0, \qquad \mathbf{x} \in \partial D, \quad t \geq 0. \tag{6.16}$$

After solving for ϕ, equation (6.14) determines p up to the function $f(t)$.

Proposition (Irrotational Flow in a Bounded Fixed Cavity)

If D is bounded and connected, the only solution of (6.15)–(6.16) *is spatially constant. Hence* $\mathbf{v} = \mathbf{0}$.

Proof.

Multiply $\Delta^{\mathrm{x}}\phi = 0$ by ϕ and integrate over D:

$$\begin{aligned} 0 &= \int_D \phi \Delta^{\mathrm{x}} \phi \, \mathrm{d}V \\ &= \int_{\partial D} \phi \nabla^{\mathrm{x}} \phi \cdot \mathbf{n} \, \mathrm{d}A - \int_D |\nabla^{\mathrm{x}} \phi|^2 \, \mathrm{d}V \\ &= - \int_D |\nabla^{\mathrm{x}} \phi|^2 \, \mathrm{d}V. \end{aligned}$$

Therefore $\nabla^{\mathrm{x}} \phi = \mathbf{0}$, so $\mathbf{v} = \mathbf{0}$. □

Example (Potential Flow in a Half-Space)

Let $D = \{\mathbf{x} \in \mathbb{E}^3 : x_3 > 0\}$ and $a > 0$. Choose the steady potential

$$\phi(\mathbf{x}) = \frac{a}{2}(x_1^2 - x_3^2).$$

Then

$$\Delta^{\mathrm{x}}\phi = a + 0 - a = 0, \qquad \mathbf{v} = \nabla^{\mathrm{x}}\phi = ax_1\mathbf{e}_1 - ax_3\mathbf{e}_3.$$

On the plane wall $x_3 = 0$, the outward normal is $\mathbf{n} = -\mathbf{e}_3$, so

$$\nabla^{\mathrm{x}}\phi \cdot \mathbf{n} = \mathbf{v} \cdot \mathbf{n} = ax_3 = 0.$$

Thus this is a nonuniform, incompressible, irrotational flow satisfying impermeability. The half-space is unbounded, so the bounded-cavity proposition does not apply.

Example (Half-Space Flow: Pressure and Particle Paths)

With no body force, steady Bernoulli gives

$$\frac{1}{2}|\mathbf{v}|^2 + \rho_0^{-1} p = \rho_0^{-1} p_s.$$

Since $|\mathbf{v}|^2 = a^2(x_1^2 + x_3^2)$,

$$p(\mathbf{x}) = p_s - \frac{\rho_0 a^2}{2}(x_1^2 + x_3^2).$$

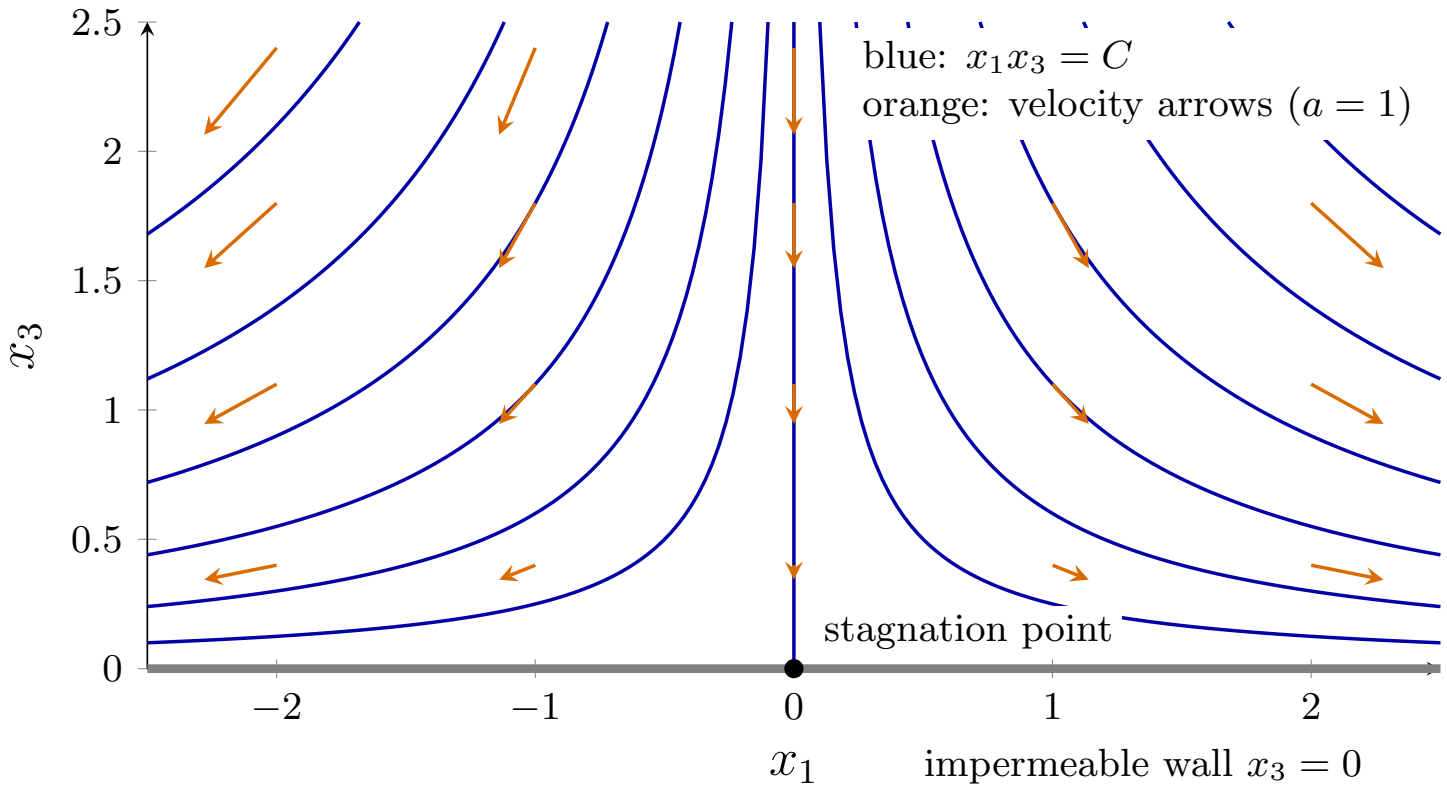
Here p_s is the pressure on the stagnation line $x_1 = x_3 = 0$.

The particle equations and their solutions are

$$\begin{aligned} \dot{x}_1 = a x_1 \quad &\Longrightarrow \quad x_1(t) = X_1 e^{at}, \\ \dot{x}_2 = 0 \quad &\Longrightarrow \quad x_2(t) = X_2, \\ \dot{x}_3 = -a x_3 \quad &\Longrightarrow \quad x_3(t) = X_3 e^{-at}. \end{aligned}$$

Particles approach the wall and move away from $x_1 = 0$.

Half-Space Potential Flow in the x_1–x_3 Plane



Since $x_1(t)x_3(t) = X_1X_3$, the streamlines are $x_1x_3 = C$. The flow is steady, so these are also the particle paths.

The stagnation line appears as the origin in this cross-section. A particle with $X_3 > 0$ approaches the wall as $t \to \infty$.



Elastic Fluids

Ideal and Elastic Fluids



Both models are inviscid and have spherical Cauchy stress:

$$S = -pI.$$

For an ideal fluid, incompressibility supplies a constraint and p is its unknown multiplier. For an elastic fluid, the density may vary and an equation of state determines pressure from density:

$$p = \pi(\rho).$$

Thus the ideal-fluid unknowns are $(\mathbf{v}, p)$, while the elastic-fluid unknowns are $(\mathbf{v}, \rho)$.

Definition (Elastic Fluid)

A continuum body with reference configuration B is an *elastic fluid* if

1. the Cauchy stress is spherical,

$$S(\mathbf{x}, t) = -p(\mathbf{x}, t) I;$$

2. pressure and spatial mass density are related by

$$p(\mathbf{x}, t) = \pi(\rho(\mathbf{x}, t)),$$

where $\pi : \mathbb{R}_+ \to \mathbb{R}$ is prescribed and

$$\pi'(s) > 0, \qquad s > 0.$$

The constitutive relation $p = \pi(\rho)$ is called an *equation of state*.

Example (A Linear Equation of State)

Fix a reference density $\rho^* > 0$, a reference pressure p^*, and a constant $c > 0$. Consider

$$\pi(\rho) = p^* + c^2(\rho - \rho^*).$$

Then

$$\pi'(\rho) = c^2 > 0.$$

A density increment $\rho - \rho^*$ produces the pressure increment

$$p - p^* = c^2(\rho - \rho^*).$$

The constant c will later appear as the linearized wave speed.

Derivation of the Elastic Fluid Equations

Balance of linear momentum and conservation of mass give

$$\rho\dot{\mathbf{v}} = \nabla^{\mathrm{x}} \cdot (-p\boldsymbol{I}) + \rho\mathbf{b}, \qquad \frac{\partial\rho}{\partial t} + \nabla^{\mathrm{x}} \cdot (\rho\mathbf{v}) = 0.$$

Since $p = \pi(\rho)$,

$$\begin{aligned} \nabla^{\mathrm{x}} \cdot (-p\boldsymbol{I}) &= -\nabla^{\mathrm{x}} p \\ &= -\nabla^{\mathrm{x}}[\pi(\rho)] \\ &= -\pi'(\rho)\nabla^{\mathrm{x}}\rho. \end{aligned}$$

Also,

$$\dot{\mathbf{v}} = \frac{\partial\mathbf{v}}{\partial t} + (\nabla^{\mathrm{x}}\mathbf{v})\mathbf{v}.$$

Elastic Fluid Equations

The spatial velocity and density satisfy

$$\rho\left[\frac{\partial \mathbf{v}}{\partial t}+(\nabla^{\mathrm{x}}\mathbf{v})\mathbf{v}\right]=-\pi'(\rho)\nabla^{\mathrm{x}}\rho+\rho\mathbf{b},$$
$$\frac{\partial \rho}{\partial t}+\nabla^{\mathrm{x}}\cdot(\rho\mathbf{v})=0, \qquad \mathbf{x}\in B_t, \quad t\geq 0. \tag{6.17}$$

There is no incompressibility constraint and no separate pressure unknown. Nevertheless, $S=-pI$ still implies that the fluid supports no shear traction.



Proposition (Frame Indifference of the Elastic-Fluid Model)

The constitutive equation

$$S(\mathbf{x}, t) = -\pi(\rho(\mathbf{x}, t))I \tag{6.18}$$

is frame indifferent.

Proof.

Fix t and consider a superposed rigid motion

$$\mathbf{x}^* = Q(t)\mathbf{x} + \mathbf{c}(t), \qquad Q^{\mathsf{T}} Q = I.$$

Let P be any region occupied by fluid, and let P^* be its image under this change of observer. Both observers assign the same mass to this fluid:

$$\int_P \rho(\mathbf{x}, t)\, \mathrm{d}V = \int_{P^*} \rho^*(\mathbf{x}^*, t)\, \mathrm{d}V^*.$$

A rigid change of observer preserves volume, since

$$\mathrm{d}V^* = |\det Q(t)|\, \mathrm{d}V = \mathrm{d}V.$$

Changing variables in the right-hand integral therefore gives

$$\int_P \rho(\mathbf{x}, t)\, \mathrm{d}V = \int_P \rho^*(Q(t)\mathbf{x} + \mathbf{c}(t), t)\, \mathrm{d}V.$$

Proof (Cont.)

Since P is arbitrary, localization gives

$$\rho^*(Q(t)\mathbf{x} + \mathbf{c}(t), t) = \rho(\mathbf{x}, t),$$

or $\rho^*(\mathbf{x}^*, t) = \rho(\mathbf{x}, t)$ at corresponding points. This is what it means for mass density to be *objective*.

Consequently, using $QIQ^{\mathsf{T}} = I$,

$$\begin{aligned} S^*(\mathbf{x}^*, t) &= -\pi(\rho^*(\mathbf{x}^*, t))I \\ &= -\pi(\rho(\mathbf{x}, t))QIQ^{\mathsf{T}} \\ &= Q[-\pi(\rho(\mathbf{x}, t))I]Q^{\mathsf{T}} \\ &= QS(\mathbf{x}, t)Q^{\mathsf{T}}. \end{aligned}$$

Thus the constitutive equation has the required transformation law. □

Elastic-Fluid Stress Power



For an elastic fluid,

$$\begin{aligned} S : L &= -\pi(\rho) I : \text{sym}(\nabla^{\text{x}} \mathbf{v}) \\ &= -\pi(\rho)\, \text{tr}(\nabla^{\text{x}} \mathbf{v}) \\ &= -\pi(\rho) \nabla^{\text{x}} \cdot \mathbf{v}. \end{aligned}$$



Proposition (Elastic-Fluid Stress Power as a Time Derivative)

For every elastic-fluid motion,

$$(\det F) S_m : L_m = -\pi \left(\frac{\rho_0}{\det F} \right) \frac{\partial}{\partial t} (\det F). \tag{6.19}$$

If $g(\mathbf{X}, s)$ *satisfies, with* $\mathbf{X}$ *held fixed,*

$$\frac{\partial g}{\partial s}(\mathbf{X}, s) = -\pi \left(\frac{\rho_0(\mathbf{X})}{s} \right),$$

then

$$(\det F) S_m : L_m = \frac{\partial}{\partial t} g(\mathbf{X}, \det F).$$

Proof.

In material variables,

$$S_m : L_m = -\pi(\rho_m)(\nabla^{\mathrm{x}} \cdot \mathbf{v})_m.$$

The volume-change and mass-conservation identities are

$$(\nabla^{\mathrm{x}} \cdot \mathbf{v})_m \det F = \frac{\partial}{\partial t} \det F, \qquad \rho_m = \frac{\rho_0}{\det F},$$

Substituting these identities gives

$$\begin{aligned}(\det F) S_m : L_m &= -\pi(\rho_m)(\det F)(\nabla^{\mathrm{x}} \cdot \mathbf{v})_m \\ &= -\pi\left(\frac{\rho_0}{\det F}\right) \frac{\partial}{\partial t}(\det F),\end{aligned}$$

which is (6.19).

Proof (Cont.)

By the chain rule, with $\mathbf{X}$ fixed and $s = \det F(\mathbf{X}, t)$,

$$\begin{aligned}\frac{\partial}{\partial t} g(\mathbf{X}, \det F) &= \frac{\partial g}{\partial s}(\mathbf{X}, \det F) \frac{\partial}{\partial t}(\det F) \\ &= -\pi \left(\frac{\rho_0}{\det F} \right) \frac{\partial}{\partial t}(\det F).\end{aligned}$$



Proposition (Mechanical Energy Inequality for an Elastic Fluid)

Every equation of state $p = \pi(\rho)$ for an elastic fluid satisfies the mechanical energy inequality for closed isothermal processes. In fact, for every such process on $[t_0, t_1]$,

$$\int_{t_0}^{t_1} \det F(\mathbf{X}, t) S_m(\mathbf{X}, t) : L_m(\mathbf{X}, t) \, \mathrm{d}t = 0 \qquad \textit{for every } \mathbf{X} \in B.$$

Proof.

By the preceding stress-power proposition,

$$(\det F) S_m : L_m = \frac{\partial}{\partial t} g(\mathbf{X}, \det F).$$

Integrating over $[t_0, t_1]$ gives

$$\begin{aligned} \int_{t_0}^{t_1} \det F(\mathbf{X}, t) S_m(\mathbf{X}, t) : L_m(\mathbf{X}, t) \, \mathrm{d}t \\ = g(\mathbf{X}, \det F(\mathbf{X}, t_1)) - g(\mathbf{X}, \det F(\mathbf{X}, t_0)). \end{aligned} \tag{6.20}$$

For a closed process, the endpoint configurations agree for all $\mathbf{X}$:

$$\boldsymbol{\phi}(\cdot, t_1) = \boldsymbol{\phi}(\cdot, t_0) \quad \Longrightarrow \quad F(\mathbf{X}, t_1) = F(\mathbf{X}, t_0).$$

Their determinants therefore agree, so the right-hand side of (6.20) vanishes. The mechanical energy inequality holds with equality. □



Elastic-Fluid Equations and Motion

A Fixed-Domain Elastic-Fluid Problem

Given π, $\mathbf{b}$, $\mathbf{v}_0$, and ρ_0, find $\mathbf{v} : D \times [0, T] \to \mathcal{V}$ and $\rho : D \times [0, T] \to \mathbb{R}$ such that

$$\left.\begin{aligned} \rho\left[\frac{\partial \mathbf{v}}{\partial t} + (\nabla^{\mathrm{x}}\mathbf{v})\mathbf{v}\right] &= -\pi'(\rho)\nabla^{\mathrm{x}}\rho + \rho\mathbf{b}, \\ \frac{\partial \rho}{\partial t} + \nabla^{\mathrm{x}} \cdot (\rho\mathbf{v}) &= 0, \\ \mathbf{v} \cdot \mathbf{n} &= 0, \\ \mathbf{v}(\cdot, 0) &= \mathbf{v}_0, \\ \rho(\cdot, 0) &= \rho_0. \end{aligned}\right\} \tag{6.21}$$

The first two equations hold in $D \times [0, T]$, the third on $\partial D \times [0, T]$, and the last two in D.

Remarks on the Elastic-Fluid Problem



The initial velocity must satisfy the wall compatibility condition

$$\mathbf{v}_0 \cdot \mathbf{n} = 0 \qquad \text{on } \partial D.$$

Even smooth data can produce shock waves, across which $\mathbf{v}$ or ρ has a jump discontinuity. Such solutions require a weak form of the balance laws. Our calculations assume all fields are as smooth as needed.

Definition (The Functions γ and ζ)

Fix $a > 0$ and define

$$\gamma(s) = \int_a^s \frac{\pi'(\xi)}{\xi}\, \mathrm{d}\xi, \qquad s > 0.$$

Then

$$\gamma'(s) = \frac{\pi'(s)}{s} > 0. \tag{6.22}$$

Thus γ is strictly increasing and has an inverse, denoted by ζ:

$$\zeta(\gamma(s)) = s.$$

Example (γ for a Linear Equation of State)

For

$$\pi(\rho) = p^* + c^2(\rho - \rho^*), \qquad \pi'(\rho) = c^2,$$

we have

$$\begin{aligned}\gamma(s) &= \int_a^s \frac{c^2}{\xi}\, \mathrm{d}\xi \\ &= c^2 \log\left(\frac{s}{a}\right).\end{aligned}$$

Solving $r = \gamma(s)$ for s gives

$$\zeta(r) = a \exp\left(\frac{r}{c^2}\right).$$



Result 6.6 (Criterion for Irrotational Motion in Elastic Fluids)

Let $\boldsymbol{\phi} : B \times [0, \infty) \to \mathbb{E}^3$ be a motion of an elastic fluid with spatial velocity $\mathbf{v}$ and mass density ρ. Suppose the body force is conservative, $\mathbf{b} = -\nabla^{\times}\beta$. If $\mathbf{v}$ is irrotational at $t = 0$, then it is irrotational for every $t \geq 0$.

Proof.

Divide the momentum equation in (6.17) by $\rho > 0$:

$$\frac{\partial \mathbf{v}}{\partial t} + (\nabla^{\times}\mathbf{v})\mathbf{v} = -\frac{\pi'(\rho)}{\rho}\nabla^{\times}\rho + \mathbf{b}.$$

Using (6.22) and $\mathbf{b} = -\nabla^{\times}\beta$ gives

$$\frac{\partial \mathbf{v}}{\partial t} + (\nabla^{\times}\mathbf{v})\mathbf{v} = -\nabla^{\times}\gamma(\rho) - \nabla^{\times}\beta. \tag{6.23}$$

Proof (Cont.)

Let $\mathbf{w} = \nabla^{\mathrm{x}} \times \mathbf{v}$. Recall that Result 6.3 gives

$$\begin{aligned}\dot{\mathbf{v}} &= \frac{\partial \mathbf{v}}{\partial t} + (\nabla^{\mathrm{x}}\mathbf{v})\mathbf{v} \\ &= \frac{\partial \mathbf{v}}{\partial t} + \frac{1}{2}\nabla^{\mathrm{x}}|\mathbf{v}|^2 + \mathbf{w} \times \mathbf{v}.\end{aligned}$$

Substituting into (6.23), we obtain

$$\frac{\partial \mathbf{v}}{\partial t} + \frac{1}{2}\nabla^{\mathrm{x}}|\mathbf{v}|^2 + \mathbf{w} \times \mathbf{v} = -\nabla^{\mathrm{x}}\gamma(\rho) - \nabla^{\mathrm{x}}\beta.$$

Taking the curl, all gradient terms vanish. Since $\nabla^{\mathrm{x}} \times \partial_t \mathbf{v} = \partial_t \mathbf{w}$, this gives

$$\frac{\partial \mathbf{w}}{\partial t} + \nabla^{\mathrm{x}} \times (\mathbf{w} \times \mathbf{v}) = \mathbf{0}. \tag{6.24}$$

Proof (Cont.)

Expanding (6.24) and using $\nabla^{\mathrm{x}} \cdot \mathbf{w} = 0$ gives

$$\dot{\mathbf{w}} - (\nabla^{\mathrm{x}}\mathbf{v})\mathbf{w} + (\nabla^{\mathrm{x}} \cdot \mathbf{v})\mathbf{w} = \mathbf{0}.$$

As in the proof of Result 6.4, define

$$\boldsymbol{\xi} = F^{-1}\mathbf{w}_m, \qquad \text{so} \qquad \mathbf{w}_m = F\boldsymbol{\xi}.$$

Differentiate at fixed $\mathbf{X}$:

$$\frac{\partial \mathbf{w}_m}{\partial t} = \frac{\partial F}{\partial t}\boldsymbol{\xi} + F\frac{\partial \boldsymbol{\xi}}{\partial t}.$$

Substitute $\boldsymbol{\xi} = F^{-1}\mathbf{w}_m$ and solve for its time derivative:

$$\frac{\partial \boldsymbol{\xi}}{\partial t} = F^{-1}\left[\frac{\partial \mathbf{w}_m}{\partial t} - \frac{\partial F}{\partial t}F^{-1}\mathbf{w}_m\right].$$

Proof (Cont.)

Recall that

$$\frac{\partial \mathbf{w}_m}{\partial t} = (\dot{\mathbf{w}})_m, \qquad \frac{\partial F}{\partial t} F^{-1} = (\nabla^{\times} \mathbf{v})_m.$$

Consequently,

$$\begin{aligned} \frac{\partial \boldsymbol{\xi}}{\partial t} &= F^{-1} \left[(\dot{\mathbf{w}})_m - (\nabla^{\times} \mathbf{v})_m \mathbf{w}_m \right] \\ &= F^{-1} \left[\dot{\mathbf{w}} - (\nabla^{\times} \mathbf{v}) \mathbf{w} \right]_m . \end{aligned}$$

The elastic-fluid vorticity equation gives

$$\dot{\mathbf{w}} - (\nabla^{\times} \mathbf{v}) \mathbf{w} = -(\nabla^{\times} \cdot \mathbf{v}) \mathbf{w}.$$

Unlike the ideal-fluid case, the divergence need not vanish. Thus

$$\begin{aligned} \frac{\partial \boldsymbol{\xi}}{\partial t} &= -F^{-1} (\nabla^{\times} \cdot \mathbf{v})_m \mathbf{w}_m \\ &= -(\nabla^{\times} \cdot \mathbf{v})_m F^{-1} \mathbf{w}_m = -(\nabla^{\times} \cdot \mathbf{v})_m \boldsymbol{\xi}. \end{aligned}$$

Proof (Cont.)

For each fixed $\mathbf{X}$, we have the linear ordinary differential equation

$$\frac{\partial \boldsymbol{\xi}}{\partial t}(\mathbf{X}, t) = -(\nabla^{\mathrm{x}} \cdot \mathbf{v})_m(\mathbf{X}, t)\boldsymbol{\xi}(\mathbf{X}, t).$$

Initial irrotationality gives $\mathbf{w}_m(\mathbf{X}, 0) = \mathbf{0}$, and hence

$$\boldsymbol{\xi}(\mathbf{X}, 0) = F^{-1}(\mathbf{X}, 0)\mathbf{w}_m(\mathbf{X}, 0) = \mathbf{0}.$$

The zero function solves this initial-value problem. By uniqueness,

$$\boldsymbol{\xi}(\mathbf{X}, t) = \mathbf{0} \quad \Longrightarrow \quad \mathbf{w}_m(\mathbf{X}, t) = F(\mathbf{X}, t)\boldsymbol{\xi}(\mathbf{X}, t) = \mathbf{0}.$$

Thus the motion remains irrotational for all time. □

Result 6.7 (Bernoulli's Theorem for Elastic Fluids)

Let an elastic-fluid motion be irrotational with $\mathbf{v} = \nabla^{\times}\phi$, and suppose $\mathbf{b} = -\nabla^{\times}\beta$. Then balance of linear momentum implies

$$\nabla^{\times}\left(\frac{\partial\phi}{\partial t} + \frac{1}{2}|\mathbf{v}|^2 + \gamma(\rho) + \beta\right) = \mathbf{0}, \tag{6.25}$$

or, equivalently,

$$\frac{\partial\phi}{\partial t} + \frac{1}{2}|\mathbf{v}|^2 + \gamma(\rho) + \beta = f(t). \tag{6.26}$$



Proof.

Under the stated assumptions, equation $(6.17)_1$ becomes

$$\rho\left[\frac{\partial}{\partial t}(\nabla^{\mathrm{x}}\phi)+\frac{1}{2}\nabla^{\mathrm{x}}|\mathbf{v}|^2\right]=-\pi'(\rho)\nabla^{\mathrm{x}}\rho-\rho\nabla^{\mathrm{x}}\beta.$$

Divide by $\rho>0$ and use (6.22):

$$\frac{\partial}{\partial t}(\nabla^{\mathrm{x}}\phi)+\frac{1}{2}\nabla^{\mathrm{x}}|\mathbf{v}|^2=-\nabla^{\mathrm{x}}\gamma(\rho)-\nabla^{\mathrm{x}}\beta.$$

Commuting $\partial/\partial t$ with ∇^{x} gives (6.25). Spatial connectedness then gives (6.26). □

Irrotational Elastic-Fluid Equations

In a fixed region D, set $\mathbf{v} = \nabla^{\mathrm{x}}\phi$. Conservation of mass becomes

$$\frac{\partial \rho}{\partial t} + \nabla^{\mathrm{x}} \cdot (\rho \nabla^{\mathrm{x}} \phi) = 0, \qquad \mathbf{x} \in D, \quad t \geq 0, \tag{6.27}$$

and impermeability becomes

$$\nabla^{\mathrm{x}} \phi \cdot \mathbf{n} = 0, \qquad \mathbf{x} \in \partial D, \quad t \geq 0. \tag{6.28}$$

Bernoulli's equation and the inverse ζ give

$$\rho = \zeta \left(f - \frac{\partial \phi}{\partial t} - \frac{1}{2} |\nabla^{\mathrm{x}} \phi|^2 - \beta \right). \tag{6.29}$$

Steady Irrotational Elastic Flow

When all fields are steady, equations (6.27) and (6.29) reduce to

$$\nabla^{\mathrm{x}} \cdot (\rho \nabla^{\mathrm{x}} \phi) = 0, \qquad \mathbf{x} \in D, \tag{6.30}$$

and

$$\rho = \zeta \left(f - \frac{1}{2} |\nabla^{\mathrm{x}} \phi|^2 - \beta \right). \tag{6.31}$$

At a reference point $\mathbf{x}^*$,

$$f = \frac{1}{2} |\mathbf{v}^*|^2 + \gamma(\rho^*) + \beta^*.$$

Equations (6.28), (6.30), and (6.31) form a nonlinear boundary-value problem for ϕ.